

Adigator

Complete Website Documentation

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Adigator — Complete Website Documentation

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1. Executive Summary

Adigator is an AI-powered advertising creative intelligence platform for performance marketers, agencies, and in-house ad teams. It provides an end-to-end workflow to configure campaigns, upload and validate display creatives, run strategic AI analysis, preview ads in realistic placement contexts, export stakeholder reports, and collaborate on creative review.

Core Capabilities

Capability	Description
Creative Validation	Catch wrong dimensions, file weight, format, and inventory mismatches before launch
Strategic Analysis	AI + rule-based scoring for goal/vertical alignment, attention path, and business impact
Preview Studio	26+ platform-native placement templates with safe-zone and crop simulation
Campaign Readiness	URL health, UTM checks, duplicate detection, and readiness scoring
Multi-Tenant Orgs	Organizations, teams, and role-based member management
Collaboration	Chat-based creative review with Brand Owner and Servicing Team roles
Reporting	PPTX strategic deck export and PDF analysis reports
Enterprise Admin	Super admin console for users, analytics, audit, health, and org provisioning

Primary User Entry Points

Entry	URL	Audience
Marketing Landing	/	Prospects
Interactive Demo	/preview-tool?demo=1&step=1	Guests (one free session)
Preview Tool (Full)	/preview-tool	Authenticated users
User Dashboard	/dashboard	Logged-in users
Sign In	/login	All users

2. Product Purpose & Value Proposition

Adigator solves the gap between creative production and campaign launch. Teams upload display creatives and receive:

1. **Technical validation** against Google Ads, Meta Ads, and Programmatic size/inventory matrices
2. **Strategic intelligence** on whether creatives align with campaign goals and industry verticals
3. **Contextual previews** showing how ads appear in Search, Feed, Stories, publisher sites, and more
4. **Actionable recommendations** prioritized by business impact
5. **Exportable reports** for client presentations and internal QA sign-off

Target Users

- Performance marketing teams
- Creative strategists and designers
- Ad operations specialists
- Digital agencies
- In-house brand marketing teams
- Enterprise organizations with multiple teams

Trust & Compliance Messaging

The marketing site displays SOC 2, GDPR, and ISO compliance badges as trust signals for enterprise buyers.

3. Technology Stack

Frontend

Technology	Version	Role
Next.js	16.2.3	App Router, SSR/SSG, API routes
React	19.2.4	UI components
TypeScript	5.9.x	Type-safe API, validation, types
Tailwind CSS	4.2.x	Styling and responsive design
Framer Motion	12.x	Page transitions, card animations
GSAP	3.15.x	Marketing page motion
Lucide React	1.8.x	Icon system
Recharts	3.8.x	Dashboard and admin charts

Backend & Services

Technology	Role
Next.js API Routes	Serverless REST endpoints
OpenAI API	Creative extraction, strategic analysis, vision element detection
Supabase	Auth, PostgreSQL, Row Level Security, file storage
Sharp	Server-side image processing
IndexedDB	Client-side creative blob storage
Socket.IO + Redis	Optional realtime presence (admin)

Export & Media Libraries

Library	Role
pptxgenjs	PPTX strategic report export
html2canvas	Preview PNG download from Preview Studio
jspdf	PDF report generation from analysis

4. System Architecture



Analysis Pipeline

1. **Upload** — Read pixel dimensions from file headers (PNG/JPEG/GIF/WebP)
 2. **Validate** — Match against platform size matrix; score inventory, IAB, DSP, auction readiness
 3. **Analyze** — OpenAI vision extraction + deterministic rule layers → strategic JSON
 4. **Preview** — Template-based placement previews + AI element detection for safe zones
 5. **Export** — Aggregate results into PPTX deck or PDF report
-

5. Complete Site Map — Every Page

Public / Marketing Pages

Route	Page Name	Description
/	Home / Landing	Hero, live animated cards, pipeline engine, feature grid, workflow modules, partner badges, CTAs
/solutions	Solutions	Tabbed content: Use Cases, Teams, Industry verticals
/about	About	Company values, stats, platform pillars, origin timeline
/product	Product	Extended marketing: Prism (performance), Adroom (creation), testimonials, blog teasers
/demo	Demo Walkthrough	Product video player, 4-step explanation, CTAs to interactive demo or sign-in
/login	Login / Register	Email/password auth, role selection on register
/login?mode=register	Registration	Username, email, password, role (Brand Owner / Servicing Team)
/preview-tool	Preview Tool (Gated)	Full 4-step Creative Studio with guest demo limits
/preview-tool?demo=1&step=1	Guest Demo Entry	One free session per browser
/preview-tool?step=1	Fresh Campaign	Always starts at Step 1 from dashboard links
/preview	Preview Tool (Direct)	Same tool without demo gate; "Live Preview" nav entry

Authenticated Application Pages

Route	Page Name	Description
/dashboard	User Dashboard	Personal workspace stats, quick actions, activity analytics
/dashboard/communications	Communications	Real-time chat + creative review workflow
/settings	Account Settings	Profile (read-only), notifications placeholder, security info
/downloads	Export History	Placeholder for future export history listing

Super Admin Console (`/dashboard/admin/\*`)

Requires platform admin role (`admin` or `super_admin` in profiles).

Route	Page Name	Description
/dashboard/admin	Platform Overview	Users, online count, DAU/MAU, retention, session metrics, charts
/dashboard/admin/organizations	Organizations	Create and list organizations (create requires super_admin)
/dashboard/admin/users	User Management	Paginated user list with search and status badges
/dashboard/admin/users/[id]	User Detail	Profile, suspend/ban, reset password, permissions
/dashboard/admin/analytics	Platform Analytics	DAU/WAU/MAU, retention, churn, session charts
/dashboard/admin/activity	Activity Logs	Platform-wide user activity with search/filter
/dashboard/admin/permissions	Permissions	Role hierarchy and feature access matrix
/dashboard/admin/audit	Audit Trail	Immutable admin action log
/dashboard/admin/health	System Health	Database, Redis, server memory checks
/dashboard/admin/notifications	Notifications	Admin notification center (placeholder)
/dashboard/admin/settings	Admin Settings	Environment variable status display

Organization Admin Console (`/dashboard/organization/\*`)

Requires org admin role (owner or org_admin in organization_members).

Route	Page Name	Description
/dashboard/organization	Org Overview	Teams, members, analyses, previews, downloads, recent activity
/dashboard/organization/teams	Teams	Create and list teams within the organization
/dashboard/organization/users	Members	Member list with roles and team assignment
/dashboard/organization/activity	Org Activity	Organization-scoped activity feed

Sidebar Links Without Pages (Planned)

Link	Status
/projects	Not implemented — linked in sidebar only
/intelligence	Not implemented — linked in sidebar only

6. Marketing Website

Navigation (`app/lib/siteNavigation.ts`)

Header links: Solutions, About

CTAs: Get a Demo → `/preview-tool?demo=1&step=1` ; Watch Demo → `/demo` ; Sign In → `/login`

Footer columns: Products, Solutions, Resources, Company

Home Page (`/`)

Positioning: Creative intelligence platform for analysis, validation, preview, and reporting.

Sections:

- Hero with rotating headline and animated `HeroLiveCards` component
- `PipelineCoreEngine` — hub-and-spoke visualization of 7 workflow modules
- Feature grid: AI creative intelligence, cross-platform compliance, preview studio
- Workflow modules: Upload & Validate → Strategic Analysis → Contextual Preview → Export & Report
- Audience insights for agencies, brands, and teams
- Trust badges (SOC 2, GDPR, ISO)
- CTAs to sign in and start demo

Solutions Page (`/solutions`)

Tabs: Use Cases (`#use-case`), Teams (`#team`), Industry (`#industry`)

Section	Content
Use Cases	Pre-launch validation, creative QA & approval, multi-platform scaling, strategic reporting
Teams	Performance marketing, creative/design, agencies, brand/in-house
Industries	Retail, healthcare/pharma, financial services, media/entertainment

About Page (`/about`)

- Values: Clarity over chaos, preview in context, built for scale
- Stats: 10 analysis layers, 3 platforms, 4-step workflow, unlimited teams per org
- Pillars: Preview Studio, Validation Engine, AI Orchestrator, Organization Platform
- Timeline: Problem → insight → platform → multi-tenant today

Product Page (`/product`)

Extended marketing not in main nav:

- Platform overview (`#overview`)
- Ad Performance / Prism (`#prism`)
- Ad Creation / Adroom (`#adroom`)

- Testimonials carousel, blog cards, DSP logo ticker
- Final CTA to preview tool

Demo Page (`/demo`)

- Embedded product walkthrough video (`/video.mp4`)
- Four demo talking points explaining the workflow
- CTAs to interactive demo or sign-in for unlimited access

7. Authentication & User Roles

Login Page (`/login`)

Sign In:

- Email and password via Supabase `signInWithPassword`
- Post-auth redirect to `/dashboard`
- Logs `user_login` activity event
- Google OAuth button rendered (placeholder — no handler currently)

Registration (`?mode=register`):

- Fields: username, email, password, confirm password
- Role selection stored in `profiles` and auth metadata:

Role ID	Label	Purpose
<code>usa_client</code>	Brand Owner	Creates conversations, assigns creatives for review
<code>end_client</code>	Servicing Team	Reviews and approves assigned creatives

Password Reset:

- "Forgot password?" links to `/login?reset=1` — reset UI not yet implemented for users
- Super admins can trigger password reset from user detail page

Role Systems (Three Layers)

Layer	Roles	Detection
Communications	<code>admin</code> , <code>usa_client</code> , <code>end_client</code>	<code>profiles.role</code>
Platform Admin	<code>super_admin</code> , <code>admin</code> , <code>moderator</code> , <code>support</code>	<code>profiles.admin_role</code> + <code>AdminAuthContext</code>
Organization	<code>owner</code> , <code>org_admin</code> , <code>team_lead</code> , <code>member</code>	<code>organization_members.member_role</code>

Session Protection

- Dashboard layout requires Supabase session; redirects to `/login` if unauthenticated
- Admin routes enforce staff role with denial screens
- Org routes enforce org admin membership

8. Preview Tool — Complete Workflow

Main component: `app/components/PreviewTool.jsx` (~3,000 lines)

Entry points: `/preview-tool` (gated) and `/preview` (direct)

URL state: `?step=1|2|3|4` , optional `?demo=1`

Step Overview

Step	Label	Primary Function
1	Setup	Select platform, goal, vertical, campaign name
2	Upload	Upload creatives, validate, compress, fix issues
3	Analysis	Run AI strategic analyzer on valid creatives
4	Preview Studio	Placement previews, safe zones, PPTX export

State Persistence

Data	Storage
Campaign config	localStorage (platform, goal, vertical, audience, URL)
Creatives metadata	localStorage via <code>workflowStorage.js</code>
Creative binary files	IndexedDB via <code>creativeAssetStore.js</code>
Analysis session	Supabase <code>analysis_sessions</code> + localStorage session ID
Authenticated saves	Supabase <code>creatives</code> , <code>analyzer_results</code> , <code>activity_logs</code>

Navigation Rules

- Dashboard links always open `?step=1` for fresh campaign setup
- Step 1 Back (logged in) → `/dashboard` ; guests → landing page `/`
- "Start New Analysis" clears all config and creatives
- Progress bar animates between steps

Step 1: Campaign Setup

Fields:

Field	Options
Platform	Google Ads, Meta Ads, Programmatic Ads
Campaign Goal	Platform-specific objectives (Awareness, Traffic, Conversion, Leads, etc.)
Industry Vertical	17 verticals (Healthcare, Technology, E-commerce, Gaming, etc.)
Campaign Name	Optional free text
Audience Stage	Cold / Warm / Hot (backend support; defaults to cold)

Requirements to advance: Platform + goal + vertical selected

Summary cards display current selections before proceeding.

Step 2: Upload & Validation

Upload flow:

1. Drag-and-drop or file picker for JPG, PNG, GIF, WebP
2. `readImageDimensionsFromBlob()` reads true pixel dimensions
3. `validateCreativeAsset()` runs platform size matrix matching
4. Creatives stored in IndexedDB; metadata in localStorage
5. Authenticated users: `saveCreative()` persists to Supabase

Per-creative validation metrics:

Metric	Description
Size	Detected dimensions (e.g., 300×250)
Placement Type	Display, native, mobile, etc.
Device Classification	Desktop, mobile, tablet, multi
IAB Compatibility	Standard IAB banner compatibility
DSP Compatibility	Count of supported DSPs (0–7)
Inventory Availability	Premium, standard, limited, unsupported
Auction Readiness	Score 0–100 with band
Premium Placement Potential	Eligibility for premium packages
Status	PASS, WARNING, or CRITICAL

Step 2 features:

Feature	Description
Platform intelligence matrix	Supported size groups per platform
Valid vs Critical lists	Creatives split by validation severity
Per-creative edit	<code>EditCreativeModal</code> for crop/resize
Compression	Per-creative or bulk target KB compression
Auto-fix	Convert formats, compress to target size
URL validation	Landing page health + AI alignment check
Campaign readiness report	Rule-based checks via <code>/api/validate</code>
Remove creative	Deletes from IndexedDB and Supabase record

Requirements to advance: At least one uploaded creative

Step 3: Strategic Analyzer

Trigger: Manual "Start Analysis" button for valid creatives

API: Platform-specific analysis via `/api/analyze-creative` (primary orchestrator)

UI tabs:

- **Overview** — Campaign health, briefing, technical QA, URL alignment, risk scoring (`AnalyzerOverview`)
- **Creative Analysis** — Per-creative strategic breakdown (`AnalyzerCreativeSection`)

Actions:

- Export PDF report from analysis panel
- Navigate to Preview Studio when complete

Persistence: Saves to `analyzer_results` and tracks `analyzer_usage` activity for authenticated users

Step 4: Preview Studio

Component: `app/components/PreviewStudio/index.jsx`

Google Ads (8 placements): Search, Display, YouTube, App, Discover, Gmail, Maps, Shopping

Meta Ads (10 placements): Facebook Feed/Stories/Reels, Instagram Feed/Stories/Explore, Messenger, Audience Network, Threads

Programmatic (8 templates): Healthcare, Technology, Food, E-commerce, Gaming, News, OTT, Blogs

Google/Meta additional tabs:

- Safe Zone Overlay — AI element detection + visualization
- Crop Simulation — Center-crop preview across standard formats

Step 4 actions:

- Export PPTX — Strategic report deck
 - Download PNG — Per-placement preview capture
 - Start New Analysis — Reset entire workflow
 - Back to Step 3
-

9. Campaign Readiness & Validation

Architecture

```
app/types/validation.ts           - TypeScript types
app/constants/googleSpecs.ts     - Google Ads requirements
app/constants/metaSpecs.ts       - Meta Ads requirements
app/constants/programmaticSpecs.ts - Programmatic rules
app/lib/validators/reportBuilder.ts - Aggregates module results
app/lib/url/healthCheck.ts        - Landing page HTTP analysis
app/lib/url/utmValidator.ts       - UTM parameter rules
app/lib/image/duplicateDetector.ts - Perceptual duplicate detection
app/api/validate/route.ts        - POST endpoint
app/api/url-validation/route.ts   - Landing page + creative alignment
```

Readiness Score

- Starts at 100
- -15 per error, -5 per warning
- Levels: Ready (≥ 85), Review Needed (60–84), Not Ready (< 60)

Validation Layers

1. **Client** (`creativeValidation.js`) — Dimension matching, mime types, file weight, inventory tier
 2. **Server** `/api/validate` — Platform validators, duplicates, URL health, UTM, alignment
 3. **Server** `/api/url-validation` — Deeper URL + creative visual alignment (OpenAI when configured)
-

10. Strategic Analyzer (Step 3)

Endpoint

`POST /api/analyze-creative` — Accepts multipart form: image + goal, platform, vertical, audience_stage

Analysis Layers

Layer	Function
1. Extraction	OpenAI vision — headline, CTA, colors, layout, emotional cues
2. Signal Classification	CTA pressure, urgency, funnel stage inference
3. Vertical Intelligence	Product category, advertising behavior
4. Creative Type Detection	Static, carousel-like, hybrid
5. Attention Analysis	First-focus path, friction points
6. Goal Alignment	Detected vs selected campaign goal
7. Vertical Alignment	Keyword fit + AI vertical match
8. Platform Alignment	Google/Meta/Programmatic fit scoring
9. Campaign Alignment	Overall aligned / partially / misaligned
10. Strategic Recommendations	Prioritized HIGH/MEDIUM/LOW fixes
11. Business Impact	Affected metrics, consequence modeling
12. Enterprise QA	Launch readiness, compliance, inventory simulation

Output Fields (per creative)

- main_strategic_problem , attention_analysis , business_consequence
- strategic_recommendations[] , expected_improvement
- strategic_alignment_score , campaign_alignment
- goal_alignment , vertical_alignment , business_impact
- adigator_analysis , extraction_signals , ai_analysis

11. Preview Studio (Step 4)

Safe Zone APIs

Platform	Endpoint	Purpose
Google	/api/google-creative-analysis	Vision element detection for safe zones
Meta	/api/meta-creative-analysis	Vision element detection for safe zones
Programmatic	/api/programmatic-creative-analysis	IAB banner element detection

Contextual Preview Engine

`POST /api/preview-engine` — Generates landing-page/environment JSON for programmatic contextual previews using analyzer output.

Device Modes

Desktop and mobile preview modes where applicable per placement.

12. User Dashboard

Route: `/dashboard`

Layout: Sidebar + topbar shell (admin/org routes use separate shells)

Overview Section

Stat Card	Source
Total Creatives	Supabase <code>creatives</code> + activity upload history + analyzer results
Valid Creatives	Cumulative validation passes (DB + activity logs)
Invalid Creatives	Cumulative validation failures (DB + activity logs)
Platforms Used	Activity log platform usage

Note: Valid and invalid counts are cumulative — a creative fixed after initial failure may appear in both totals.

Quick Actions

Action	Destination
Open Preview Tool	<code>/preview-tool?step=1</code>
Resume Analysis	<code>/preview-tool?step=3</code>
Organization Console	<code>/dashboard/organization</code> (org admins only)
Super Admin Console	<code>/dashboard/admin</code> (platform admins only)

Activity Analytics

`UserAnalyticsCharts` component displays:

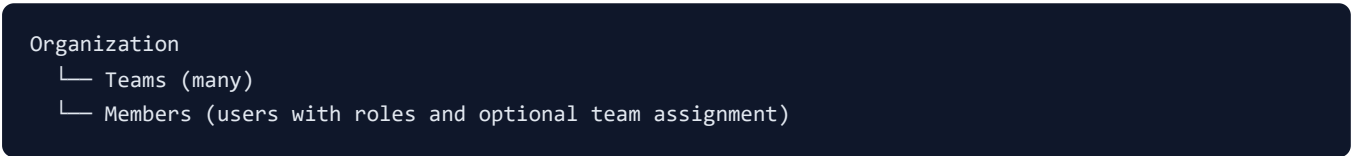
- Platform usage breakdown
- Analysis activity over time (14-day window)
- Preview studio opens, reports downloaded, analyzer runs

Data Sources

- `fetchUserCreatives()` — Direct Supabase query
- `fetchUserDashboardAnalytics()` — Merged local + remote activity logs (up to 5,000 events)
- `fetchAnalyzerResultCreativeIds()` — Creatives with saved analysis
- `computeCreativeCountStats()` — Merges all sources for accurate totals

13. Organization Platform (Multi-Tenant)

Data Model



Tables: `organizations`, `teams`, `organization_members`

Member roles: `owner`, `org_admin`, `team_lead`, `member`

Org status: `active`, `suspended`, `trial`

Organization Admin Features

Feature	Description
Org dashboard	Member count, team count, activity metrics
Team management	Create teams, view team list
Member management	View members, assign roles and teams
Activity feed	Org-scoped user activity

API Endpoints

Method	Route	Purpose
GET	<code>/api/organization/v1/membership</code>	Current user's org membership
GET	<code>/api/organization/v1/members</code>	List org members
PATCH	<code>/api/organization/v1/members/[id]</code>	Update role, team, status
GET	<code>/api/organization/v1/teams</code>	List teams
POST	<code>/api/organization/v1/teams</code>	Create team
GET	<code>/api/organization/v1/dashboard</code>	Org metrics

Setup

Run `supabase/migrations/20260616_organizations_teams.sql` Or `supabase/RUN_ORGANIZATIONS_SETUP.sql` in Supabase SQL Editor.

Super admins create organizations via `/dashboard/admin/organizations` .

14. Super Admin Platform

RBAC Roles

`super_admin` → `admin` → `moderator` → `support` → `user`

Permissions

`users:read` , `users:write` , `audit:read` , `analytics:read` , `health:read` , `permissions:write`

Admin API (`/api/admin/v1/*`)

Endpoint	Purpose
GET <code>/dashboard</code>	Platform metrics
GET <code>/analytics</code>	Extended analytics (DAU/WAU/MAU, retention)
GET <code>/health</code>	DB, Redis, realtime health
GET <code>/users</code>	Paginated user list (CSV export available)
GET <code>/users/[id]</code>	User detail
POST <code>/users/[id]/actions</code>	Suspend, ban, reset password, update, delete
GET <code>/activity</code>	Cross-user activity logs
GET <code>/audit</code>	Admin audit trail
GET <code>/permissions</code>	Role definitions and feature matrix
GET/POST <code>/organizations</code>	List/create organizations

Admin Database Tables

`admin_user_sessions` , `admin_user_activity` , `admin_feature_permissions` , `admin_audit_logs` ,
`admin_notifications` , `admin_feature_usage` , `admin_error_logs` , `admin_role_definitions`

Setup

Run `supabase/migrations/20260615_admin_enterprise_platform.sql`

Grant access: `UPDATE profiles SET role = 'admin', admin_role = 'super_admin' WHERE email = '...'`

See `docs/ADMIN_PLATFORM_SETUP.md` for full setup guide.

15. Communication Platform

Route: `/dashboard/communications`

Component: `CommunicationPlatform.jsx`

Features

- Real-time chat via Supabase realtime + `useCommunicationRealtime`
- Direct and project conversations
- Message types: text, file, creative attachment, system events
- Creative review workflow with statuses: pending → in_review → approved / revision_requested
- Activity timeline per conversation
- In-app notifications
- User presence tracking

Review Workflow

1. Brand Owner creates conversation and assigns creatives
2. Servicing Team receives assignment notification
3. Reviewer approves or requests revision with notes
4. Status updates tracked in `creative_reviews` table

API Endpoints

Method	Route	Purpose
GET/POST	<code>/api/communications/conversations</code>	List/create conversations
GET/PATCH	<code>/api/communications/conversations/[id]</code>	Detail/update
GET/POST	<code>/api/communications/conversations/[id]/messages</code>	Messages
GET/POST	<code>/api/communications/conversations/[id]/activity</code>	Activity events
POST	<code>/api/communications/attachments/[id]/review</code>	Submit review
GET/PATCH	<code>/api/communications/notifications</code>	Notifications
GET/PATCH	<code>/api/communications/users</code>	User search/update

16. Settings, Downloads & Account

Settings (`/settings`)

Section	Status
Profile	Read-only name and email from Supabase auth
Notifications	Placeholder — email alerts coming soon
Security	Password/session managed by login; link to re-authenticate

Downloads (`/downloads`)

Placeholder page directing users to export from Preview Tool Step 3 (PDF) or Step 4 (PPTX).

Admin Settings (`/dashboard/admin/settings`)

Displays environment variable configuration status (Supabase URL, keys, Realtime URL).

17. Guest Demo Access

Logic file: `app/lib/demoAccess.js`

Gate component: `PreviewToolGate.jsx`

Rules

User Type	Access
Authenticated	Unlimited Preview Tool use; data persists to Supabase
Guest (first visit)	One demo session via <code>?demo=1&step=1</code>
Guest (demo used, tab closed)	Blocked — must sign in

Storage Keys

Key	Storage	Purpose
<code>adigator_guest_demo_used</code>	localStorage	Permanent "demo consumed" flag
<code>adigator_guest_demo_active</code>	sessionStorage	Active tab session flag
<code>adigator_preview_refresh</code>	sessionStorage	Distinguishes refresh vs navigate-away

Blocked User Experience

Shows sign-in CTA and link to watch demo video instead of Preview Tool.

18. Complete API Reference

Creative Intelligence

Method	Route	Purpose
POST	/api/analyze-creative	Full strategic analysis orchestrator
POST	/api/validate	Campaign readiness report
POST	/api/url-validation	Landing page + creative alignment
POST	/api/google-creative-analysis	Google safe zone element detection
POST	/api/meta-creative-analysis	Meta safe zone element detection
POST	/api/programmatic-creative-analysis	Programmatic element detection

Preview

Method	Route	Purpose
POST	/api/preview-engine	Contextual preview engine
GET	/api/preview-engine	Engine metadata
POST	/api/preview-templates	Legacy template generation

Creatives & Sessions

Method	Route	Purpose
GET/POST	/api/creatives	List/save user creatives
POST	/api/session/create	Create analysis session
POST	/api/session/update	Update session
GET	/api/session/[id]	Fetch session

Activity

Method	Route	Purpose
POST	/api/activity/log	Log user activity
GET	/api/activity/list	List activity (admin-gated)

All routes above require Bearer token from Supabase session unless noted.

19. Database Schema (Supabase)

Core Tables

Table	Purpose
creatives	Uploaded creative metadata (name, type, size, is_valid, validation_status, file_url)
analyzer_results	JSON analysis per creative/platform/goal
activity_logs	User activity audit trail
analysis_sessions	Persistent analysis session state
user_activity_events	Structured activity events

Communication Tables

Table	Purpose
profiles	User roles, admin_role, status, company info
conversations	Chat threads (direct/project)
conversation_participants	Membership and read state
messages	Text, file, creative, system messages
attachments	File/creative attachments
creative_reviews	Review status and notes
comm_activity_events	Communication audit trail
comm_notifications	In-app notifications

Organization Tables

Table	Purpose
organizations	Org name, slug, status, plan, industry
teams	Teams within an organization
organization_members	User membership, role, team assignment

Admin Tables

Table	Purpose
admin_audit_logs	Admin actions on users/system
admin_user_sessions	Login sessions
admin_feature_permissions	Per-user feature toggles
admin_role_definitions	RBAC role catalog

Storage

Bucket: creative-assets

Path pattern: {user_id}/{creative_id}/{filename}

Migration Files

File	Purpose
20260526_adigator_core_tables_rls.sql	Core tables + RLS
20260527_communication_platform.sql	Communications schema
20260615_admin_enterprise_platform.sql	Admin enterprise tables
20260616_organizations_teams.sql	Multi-tenant org schema
RUN_PREVIEW_TOOL_TABLES.sql	Idempotent preview tool setup
RUN_ORGANIZATIONS_SETUP.sql	Org setup helper
FIX_DATABASE.sql	Schema repair script

20. Data Services & Analytics

`supabaseDataService.js`

Function	Purpose
<code>saveCreative()</code>	Upload blob to storage + POST <code>/api/creatives</code>
<code>saveAnalyzerResult()</code>	Insert analyzer_results row
<code>trackUserActivity()</code>	POST <code>/api/activity/log</code> with localStorage fallback
<code>trackValidationOutcome()</code>	Deduped validation pass/fail events
<code>fetchUserCreatives()</code>	Query user's creatives from Supabase
<code>deleteCreativeRecord()</code>	Delete creative row

Activity Event Types

`page_visit`, `upload`, `validation_outcome`, `analyzer_usage`, `navigation`, `download`, `generate_action`, `platform_selection`, `button_click`, `user_login`

Dashboard Analytics (`userDashboardAnalytics.js`)

Merges local + remote activity logs to compute:

- Analysis by day (14-day chart)
- Platform usage counts
- Preview studio opens, reports downloaded, analyzer runs
- Creative validity sets from upload and validation_outcome events

21. Export & Reporting

PPTX Export

- **Trigger:** Step 4 "Export PPTX" button
- **File:** `app/lib/exportPptx.js`
- **Library:** `pptxgenjs`
- **Content:** Campaign summary, per-creative analysis, alignment scores, recommendations
- **Filename:** `Adigator_Advertising_Intelligence_{N}Creatives.pptx`

PDF Export

- **Trigger:** Step 3 analysis panel download
- **Library:** `jspdf`

- **Content:** Analysis summary and per-creative findings

PNG Export

- **Trigger:** Preview Studio download button per placement
- **Library:** html2canvas
- **Content:** Rendered placement template capture

22. Supported Platforms, Goals & Verticals

Platforms

ID	Name	Placements	Safe Zone
google_ads	Google Ads	8	Yes
meta_ads	Meta Ads	10	Yes
programmatic	Programmatic Ads	8	No

Campaign Goals

Google & Meta: Awareness, Traffic, Conversion, Lead Generation, Engagement, App Installs, Retargeting, Video Views

Programmatic: Awareness, Consideration, Conversion

Audience Stages

Cold (first-touch), Warm (partially familiar), Hot (retargeting/high-intent)

Industry Verticals (17)

Healthcare, Technology, Automotive, News/Media, Sports, Fitness, Business/Finance, Luxury, Travel, Hotels, Restaurants/Food, Banking/FinTech, Real Estate, Education/EdTech, Gaming, Entertainment/OTT, E-commerce/Retail, Fashion

Standard Display Sizes (examples)

300×250, 336×280, 728×90, 970×90, 970×250, 160×600, 300×600, 320×50, 320×100, 468×60, and more per platform registry.

23. Environment Variables

```
# Required – Supabase
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=your-anon-key

# Required for admin/org APIs
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key

# Required for AI analysis
OPENAI_API_KEY=sk-...
OPENAI_MODEL=gpt-4o # optional, defaults to gpt-4o

# Optional – Realtime presence
NEXT_PUBLIC_REALTIME_URL=http://localhost:4010
REDIS_URL=redis://localhost:6379
REALTIME_PORT=4010

# Optional – Email alerts (admin)
SMTP_HOST=
SMTP_USER=
ADMIN_EMAIL=
```

24. Development & Deployment

Local Development

```
npm install
npm run dev # http://localhost:3000
```

Production Build

```
npm run build
npm start
```

Lint

```
npm run lint
```

Deployment

- **Platform:** Vercel (Next.js native)
- **Production URL:** adigator.in
- **Database:** Apply Supabase migrations via SQL Editor
- **Realtime server:** Optional separate service at `services/realtime-server/`

Database Setup Order

1. `RUN_PREVIEW_TOOL_TABLES.sql` or core migrations
 2. `RUN_COMMUNICATION_PLATFORM.sql`
 3. `20260615_admin_enterprise_platform.sql`
 4. `20260616_organizations_teams.sql` or `RUN_ORGANIZATIONS_SETUP.sql`
-

25. Project Structure

```
Adigator/
├─ app/
│   ├─ api/                # 37+ API route handlers
│   ├─ components/
│   │   ├─ PreviewTool.jsx    # Main 4-step wizard
│   │   ├─ PreviewToolGate.jsx # Guest demo gate
│   │   ├─ PreviewStudio/     # Step 4 previews
│   │   ├─ communications/    # Chat/review platform
│   │   ├─ admin/            # Super admin UI shell
│   │   ├─ organization/      # Org admin UI shell
│   │   ├─ marketing/         # Landing page components
│   │   ├─ dashboard/         # User dashboard charts
│   │   └─ Validation*.tsx     # Readiness UI
│   ├─ dashboard/           # All dashboard pages
│   ├─ login/               # Auth pages
│   ├─ lib/                 # Business logic, services, validators
│   ├─ preview-tool/        # Gated preview tool route
│   ├─ preview/             # Direct preview tool route
│   ├─ settings/            # Account settings
│   └─ styles/              # login.css, premium-ui.css
├─ docs/                   # Documentation (this file)
├─ supabase/               # SQL migrations and setup scripts
├─ services/realtime-server/ # Optional Socket.IO presence server
├─ public/                 # Static assets
└─ package.json
```

26. Known Limitations & Roadmap Items

Item	Status
Google OAuth on login	UI only — no handler
User password reset	Link exists; reset flow not implemented
/projects page	Sidebar link only
/intelligence page	Sidebar link only
User notification settings	Placeholder on /settings
Admin notifications	Placeholder on admin console
Downloads history	Placeholder on /downloads
Audience stage UI	Backend support; no visible Step 1 selector
Guest demo enforcement	Client-side only (no server-side limit)

27. Appendix: Key Design Decisions

Hybrid AI + Rules

OpenAI handles vision extraction and strategic language; deterministic rules enforce platform size matrices and scoring consistency.

IndexedDB for Creative Blobs

Full-resolution images exceed localStorage limits; IndexedDB enables refresh-safe re-hydration.

Template-Based Preview Studio

26 hardcoded placement templates for speed and consistency; OpenAI used only for safe-zone detection.

Cumulative Dashboard Metrics

Valid/invalid creative counts are cumulative across upload history to reflect total testing activity, not just current state.

Multi-Tenant Architecture

Organizations provide team isolation within a single Supabase instance with RLS policies scoped per org membership.

Activity Log Privacy

User aggregate reads from `activity_logs` are admin-only; users access their own events via merged local + permitted remote reads for dashboard stats.

Document Information

Item	Value
File	<code>docs/ADIGATOR_COMPLETE_WEBSITE_DOCUMENTATION.md</code>
Companion docs	<code>docs/ADIGATOR_PLATFORM_DOCUMENTATION.md</code> , <code>docs/ADMIN_PLATFORM_SETUP.md</code>
Agent rules	<code>AGENTS.md</code>
Schema reference	<code>supabase/migrations/</code>

This document covers every page, feature, tool, API endpoint, database table, workflow, and role system in the Adigator platform as of June 2026.